

|                                      |              |             |                |                                   |                                   |                             |              |             |           |
|--------------------------------------|--------------|-------------|----------------|-----------------------------------|-----------------------------------|-----------------------------|--------------|-------------|-----------|
| Totally Integrated Automation Portal |              |             |                |                                   |                                   |                             |              |             |           |
| FB_Warehouse [FB6]                   |              |             |                |                                   |                                   |                             |              |             |           |
| FB_Warehouse Eigenschaften           |              |             |                |                                   |                                   |                             |              |             |           |
| Allgemein                            |              |             |                |                                   |                                   |                             |              |             |           |
| Name                                 | FB_Warehouse |             | Nummer         | 6                                 |                                   | Typ                         | FB           |             |           |
| Sprache                              | SCL          |             | Nummerierung   | Manuell                           |                                   |                             |              |             |           |
| Information                          |              |             |                |                                   |                                   |                             |              |             |           |
| Titel                                |              |             | Autor          |                                   |                                   | Kommentar                   |              |             |           |
| Familie                              |              |             | Version        | 0.1                               |                                   | Anwenderdefinierte ID       |              |             |           |
| Name                                 | Datentyp     | Defaultwert | Remanenz       | Erreichbar aus HMI/OPC UA/Web API | Schreibbar aus HMI/OPC UA/Web API | Sichtbar in HMI Engineering | Einstellwert | Überwachung | Kommentar |
| ▼ Input                              |              |             |                |                                   |                                   |                             |              |             |           |
| Enable                               | Bool         | false       | Nicht remanent | True                              | True                              | True                        | False        |             |           |
| Reset                                | Bool         | false       | Nicht remanent | True                              | True                              | True                        | False        |             |           |
| LT_Crane_Feed                        | Bool         | false       | Nicht remanent | True                              | True                              | True                        | False        |             |           |
| Stacker-Crane_1_Moving_X             | Bool         | false       | Nicht remanent | True                              | True                              | True                        | False        |             |           |
| Stacker-Crane_1_Moving_Z             | Bool         | false       | Nicht remanent | True                              | True                              | True                        | False        |             |           |
| Stacker-Crane_1_Left_Limit           | Bool         | false       | Nicht remanent | True                              | True                              | True                        | False        |             |           |
| Stacker-Crane_1_Right_Limit          | Bool         | false       | Nicht remanent | True                              | True                              | True                        | False        |             |           |
| Stacker-Crane_1_Middle_Limit         | Bool         | false       | Nicht remanent | True                              | True                              | True                        | False        |             |           |
| ▼ Output                             |              |             |                |                                   |                                   |                             |              |             |           |
| Loading_to_Crane_Conveyor            | Bool         | false       | Nicht remanent | True                              | True                              | True                        | False        |             |           |
| Stacker-Crane_1_Left                 | Bool         | false       | Nicht remanent | True                              | True                              | True                        | False        |             |           |
| Stacker-Crane_1_Right                | Bool         | false       | Nicht remanent | True                              | True                              | True                        | False        |             |           |
| Stacker-Crane_1_Lift                 | Bool         | false       | Nicht remanent | True                              | True                              | True                        | False        |             |           |

| Totally Integrated Automation Portal |          |             |                |                                   |                                   |                             |              |             |  |  |           |
|--------------------------------------|----------|-------------|----------------|-----------------------------------|-----------------------------------|-----------------------------|--------------|-------------|--|--|-----------|
| Name                                 | Datentyp | Defaultwert | Remanenz       | Erreichbar aus HMI/OPC UA/Web API | Schreibbar aus HMI/OPC UA/Web API | Sichtbar in HMI Engineering | Einstellwert | Überwachung |  |  | Kommentar |
| Stacker-Crane_1_Target_1_Position    | DInt     | 0           | Nicht remanent | True                              | True                              | True                        | False        |             |  |  |           |
| Busy                                 | Bool     | false       | Nicht remanent | True                              | True                              | True                        | False        |             |  |  |           |
| Ready                                | Bool     | false       | Nicht remanent | True                              | True                              | True                        | False        |             |  |  |           |
| WarehouseFull                        | Bool     | false       | Nicht remanent | True                              | True                              | True                        | False        |             |  |  |           |
| OccupiedPositions                    | DInt     | 0           | Nicht remanent | True                              | True                              | True                        | False        |             |  |  |           |
| InOut                                |          |             |                |                                   |                                   |                             |              |             |  |  |           |
| ▼ Static                             |          |             |                |                                   |                                   |                             |              |             |  |  |           |
| State                                | Int      | 0           | Nicht remanent | True                              | True                              | True                        | False        |             |  |  |           |
| NextPosition                         | DInt     | 0           | Nicht remanent | True                              | True                              | True                        | False        |             |  |  |           |
| MovementStarted                      | Bool     | false       | Nicht remanent | True                              | True                              | True                        | False        |             |  |  |           |
| ▼ TimerLiftUp                        | TON_TIME |             | Nicht remanent | True                              | True                              | True                        | False        |             |  |  |           |
| PT                                   | Time     | T#0ms       | Nicht remanent | True                              | True                              | True                        | False        |             |  |  |           |
| ET                                   | Time     | T#0ms       | Nicht remanent | True                              | False                             | True                        | False        |             |  |  |           |
| IN                                   | Bool     | false       | Nicht remanent | True                              | True                              | True                        | False        |             |  |  |           |
| Q                                    | Bool     | false       | Nicht remanent | True                              | False                             | True                        | False        |             |  |  |           |
| ▼ TimerLiftDown                      | TON_TIME |             | Nicht remanent | True                              | True                              | True                        | False        |             |  |  |           |
| PT                                   | Time     | T#0ms       | Nicht remanent | True                              | True                              | True                        | False        |             |  |  |           |
| ET                                   | Time     | T#0ms       | Nicht remanent | True                              | False                             | True                        | False        |             |  |  |           |
| IN                                   | Bool     | false       | Nicht remanent | True                              | True                              | True                        | False        |             |  |  |           |
| Q                                    | Bool     | false       | Nicht remanent | True                              | False                             | True                        | False        |             |  |  |           |
| ▼ TimerDeposit                       | TON_TIME |             | Nicht remanent | True                              | True                              | True                        | True         |             |  |  |           |
| PT                                   | Time     | T#0ms       | Nicht remanent | True                              | True                              | True                        | False        |             |  |  |           |
| ET                                   | Time     | T#0ms       | Nicht remanent | True                              | False                             | True                        | False        |             |  |  |           |
| IN                                   | Bool     | false       | Nicht remanent | True                              | True                              | True                        | False        |             |  |  |           |
| Q                                    | Bool     | false       | Nicht remanent | True                              | False                             | True                        | False        |             |  |  |           |

| Totally Integrated Automation Portal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |             |          |                                   |                                   |                             |              |             |           |
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| Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Datentyp | Defaultwert | Remanenz | Erreichbar aus HMI/OPC UA/Web API | Schreibbar aus HMI/OPC UA/Web API | Sichtbar in HMI Engineering | Einstellwert | Überwachung | Kommentar |
| Temp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |             |          |                                   |                                   |                             |              |             |           |
| Constant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |             |          |                                   |                                   |                             |              |             |           |
| <pre>0001 // Warehouse position counters 0002 IF #State = 7 THEN 0003     #OccupiedPositions := 54; 0004     #FreePositions := 0; 0005 ELSE 0006     #OccupiedPositions := #NextPosition - 1; 0007     #FreePositions := 54 - #OccupiedPositions; 0008 END_IF; 0009 // Deposit timer 0010 #TimerDeposit ( 0011     IN := (#State = 4), 0012     PT := T#2s 0013 ); 0014 0015 // Reset 0016 IF #Reset THEN 0017     #State := 0; 0018     #NextPosition := 1; 0019     #MovementStarted := FALSE; 0020     #Loading_to_Crane_Conveyor := FALSE; 0021     #StackerCrane_1_Left := FALSE; 0022     #StackerCrane_1_Right := FALSE; 0023     #StackerCrane_1_Lift := FALSE; 0024     #StackerCrane_1_Target_1_Position := 0; 0025 ELSE 0026     // Status 0027     #Ready := (#State = 0); 0028     #Busy := (#State &gt; 0) AND (#State &lt; 7); 0029     #WarehouseFull := (#State = 7); 0030     CASE #State OF 0031         // 0 - Wait for pallet 0032         0: 0033             #StackerCrane_1_Target_1_Position := 0; 0034             IF #Enable THEN 0035                 #Loading_to_Crane_Conveyor := TRUE; 0036                 IF #LT_Crane_Feed THEN 0037                     #Loading_to_Crane_Conveyor := FALSE; 0038                     #StackerCrane_1_Left := TRUE; 0039                     #State := 1; 0040                 END_IF; 0041             END_IF; 0042         // 1 - Pick up pallet 0043         1: 0044             IF #StackerCrane_1_Left_Limit THEN 0045                 // Lift pallet</pre> |          |             |          |                                   |                                   |                             |              |             |           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |             |          |                                   |                                   |                             |              |             |           |

|                                         |                                                     |  |
|-----------------------------------------|-----------------------------------------------------|--|
| Totally Integrated<br>Automation Portal |                                                     |  |
| 0046                                    | #StackerCrane_1_Lift := TRUE;                       |  |
| 0047                                    | // Retract forks                                    |  |
| 0048                                    | #StackerCrane_1_Left := FALSE;                      |  |
| 0049                                    | #State := 2;                                        |  |
| 0050                                    | END_IF;                                             |  |
| 0051                                    | // 2 - Wait until forks reach MIDDLE                |  |
| 0052                                    | 2:                                                  |  |
| 0053                                    | IF #StackerCrane_1_Middle_Limit THEN                |  |
| 0054                                    | #StackerCrane_1_Target_1_Position := #NextPosition; |  |
| 0055                                    | #MovementStarted := FALSE;                          |  |
| 0056                                    | #State := 3;                                        |  |
| 0057                                    | END_IF;                                             |  |
| 0058                                    | // 3 - Move X/Z and extend forks RIGHT              |  |
| 0059                                    | 3:                                                  |  |
| 0060                                    | #StackerCrane_1_Target_1_Position := #NextPosition; |  |
| 0061                                    | IF #StackerCrane_1_Moving_X                         |  |
| 0062                                    | OR #StackerCrane_1_Moving_Z THEN                    |  |
| 0063                                    | #MovementStarted := TRUE;                           |  |
| 0064                                    | END_IF;                                             |  |
| 0065                                    | IF #MovementStarted                                 |  |
| 0066                                    | AND NOT #StackerCrane_1_Moving_X                    |  |
| 0067                                    | AND NOT #StackerCrane_1_Moving_Z THEN               |  |
| 0068                                    | #MovementStarted := FALSE;                          |  |
| 0069                                    | // Extend forks into rack                           |  |
| 0070                                    | #StackerCrane_1_Right := TRUE;                      |  |
| 0071                                    | END_IF;                                             |  |
| 0072                                    | IF #StackerCrane_1_Right_Limit THEN                 |  |
| 0073                                    | // Lower pallet onto rack                           |  |
| 0074                                    | // Lift remains FALSE from this point               |  |
| 0075                                    | #StackerCrane_1_Lift := FALSE;                      |  |
| 0076                                    | #State := 4;                                        |  |
| 0077                                    | END_IF;                                             |  |
| 0078                                    | // 4 - Wait for pallet to settle                    |  |
| 0079                                    | 4:                                                  |  |
| 0080                                    | #StackerCrane_1_Target_1_Position := #NextPosition; |  |
| 0081                                    | IF #TimerDeposit.Q THEN                             |  |
| 0082                                    | // Retract forks only after pallet has settled      |  |
| 0083                                    | #StackerCrane_1_Right := FALSE;                     |  |
| 0084                                    | #State := 5;                                        |  |
| 0085                                    | END_IF;                                             |  |
| 0086                                    | // 5 - Wait until forks reach MIDDLE                |  |
| 0087                                    | 5:                                                  |  |
| 0088                                    | #StackerCrane_1_Target_1_Position := #NextPosition; |  |
| 0089                                    | IF #StackerCrane_1_Middle_Limit THEN                |  |
| 0090                                    | #StackerCrane_1_Target_1_Position := 55;            |  |
| 0091                                    | #MovementStarted := FALSE;                          |  |
| 0092                                    | #State := 6;                                        |  |
| 0093                                    | END_IF;                                             |  |
| 0094                                    | // 6 - Return to REST                               |  |
| 0095                                    | 6:                                                  |  |
| 0096                                    | #StackerCrane_1_Target_1_Position := 55;            |  |
| 0097                                    | IF #StackerCrane_1_Moving_X                         |  |
| 0098                                    | OR #StackerCrane_1_Moving_Z THEN                    |  |
| 0099                                    | #MovementStarted := TRUE;                           |  |
| 0100                                    | END_IF;                                             |  |
| 0101                                    | IF #MovementStarted                                 |  |
| 0102                                    | AND NOT #StackerCrane_1_Moving_X                    |  |
| 0103                                    | AND NOT #StackerCrane_1_Moving_Z THEN               |  |
| 0104                                    | #StackerCrane_1_Target_1_Position := 0;             |  |
|                                         |                                                     |  |

```
0105      #MovementStarted := FALSE;
0106      IF #NextPosition < 54 THEN
0107          #NextPosition := #NextPosition + 1;
0108          #State := 0;
0109      ELSE
0110          #State := 7;
0111      END_IF;
0112  END_IF;
0113  // 7 - Warehouse full
0114  7:
0115      #StackerCrane_1_Target_1_Position := 0;
0116  ELSE
0117      #State := 0;
0118  END_CASE;
0119 END_IF;
```

| Symbol                                 | Adresse | Typ       | Kommentar |
|----------------------------------------|---------|-----------|-----------|
| #Busy                                  |         | Bool      |           |
| #Enable                                |         | Bool      |           |
| #Load-<br>ing_to_Crane_Conveyor        |         | Bool      |           |
| #LT_Crane_Feed                         |         | Bool      |           |
| #MovementStarted                       |         | Bool      |           |
| #NextPosition                          |         | DInt      |           |
| #OccupiedPositions                     |         | DInt      |           |
| #Ready                                 |         | Bool      |           |
| #Reset                                 |         | Bool      |           |
| #StackerCrane_1_Left                   |         | Bool      |           |
| #Stacker-<br>Crane_1_Left_Limit        |         | Bool      |           |
| #StackerCrane_1_Lift                   |         | Bool      |           |
| #StackerCrane_1_Mid-<br>dle_Limit      |         | Bool      |           |
| #StackerCrane_1_Mov-<br>ing_X          |         | Bool      |           |
| #StackerCrane_1_Mov-<br>ing_Z          |         | Bool      |           |
| #StackerCrane_1_Right                  |         | Bool      |           |
| #Stacker-<br>Crane_1_Right_Limit       |         | Bool      |           |
| #StackerCrane_1_Tar-<br>get_1_Position |         | DInt      |           |
| #State                                 |         | Int       |           |
| #TimerDeposit                          |         | IEC_Timer |           |
| #TimerDeposit.Q                        |         | Bool      |           |
| #WarehouseFull                         |         | Bool      |           |

|                                      |                 |              |             |                                    |                                    |                             |              |             |           |
|--------------------------------------|-----------------|--------------|-------------|------------------------------------|------------------------------------|-----------------------------|--------------|-------------|-----------|
| Totally Integrated Automation Portal |                 |              |             |                                    |                                    |                             |              |             |           |
| FB_Warehouse_DB [DB14]               |                 |              |             |                                    |                                    |                             |              |             |           |
| FB_Warehouse_DB Eigenschaften        |                 |              |             |                                    |                                    |                             |              |             |           |
| Allgemein                            |                 |              |             |                                    |                                    |                             |              |             |           |
| Name                                 | FB_Warehouse_DB | Nummer       | 14          | Typ                                | DB                                 |                             |              |             |           |
| Sprache                              | DB              | Nummerierung | Automatisch |                                    |                                    |                             |              |             |           |
| Information                          |                 |              |             |                                    |                                    |                             |              |             |           |
| Titel                                |                 | Autor        |             | Kommentar                          |                                    |                             |              |             |           |
| Familie                              |                 | Version      | 0.1         | Anwenderdefinierte ID              |                                    |                             |              |             |           |
| Name                                 | Datentyp        | Startwert    | Remanenz    | Erreichbar aus HMI/O PC UA/Web API | Schreibbar aus HMI/O PC UA/Web API | Sichtbar in HMI Engineering | Einstellwert | Überwachung | Kommentar |
| ▼ Input                              |                 |              |             |                                    |                                    |                             |              |             |           |
| Enable                               | Bool            | false        | False       | True                               | True                               | True                        | False        |             |           |
| Reset                                | Bool            | false        | False       | True                               | True                               | True                        | False        |             |           |
| LT_Crane_Feed                        | Bool            | false        | False       | True                               | True                               | True                        | False        |             |           |
| Stacker-Crane_1_Moving_X             | Bool            | false        | False       | True                               | True                               | True                        | False        |             |           |
| Stacker-Crane_1_Moving_Z             | Bool            | false        | False       | True                               | True                               | True                        | False        |             |           |
| Stacker-Crane_1_Left_Limit           | Bool            | false        | False       | True                               | True                               | True                        | False        |             |           |
| Stacker-Crane_1_Right_Limit          | Bool            | false        | False       | True                               | True                               | True                        | False        |             |           |
| Stacker-Crane_1_Middle_Limit         | Bool            | false        | False       | True                               | True                               | True                        | False        |             |           |
| ▼ Output                             |                 |              |             |                                    |                                    |                             |              |             |           |
| Loading_to_Crane_Conveyor            | Bool            | false        | False       | True                               | True                               | True                        | False        |             |           |
| Stacker-Crane_1_Left                 | Bool            | false        | False       | True                               | True                               | True                        | False        |             |           |
| Stacker-Crane_1_Right                | Bool            | false        | False       | True                               | True                               | True                        | False        |             |           |
| Stacker-Crane_1_Lift                 | Bool            | false        | False       | True                               | True                               | True                        | False        |             |           |
|                                      |                 |              |             |                                    |                                    |                             |              |             |           |
|                                      |                 |              |             |                                    |                                    |                             |              |             |           |

| Totally Integrated Automation Portal |          |           |          |                                    |                                    |                             |              |             |           |  |
|--------------------------------------|----------|-----------|----------|------------------------------------|------------------------------------|-----------------------------|--------------|-------------|-----------|--|
| Name                                 | Datentyp | Startwert | Remanenz | Erreichbar aus HMI/O PC UA/Web API | Schreibbar aus HMI/O PC UA/Web API | Sichtbar in HMI Engineering | Einstellwert | Überwachung | Kommentar |  |
| Stacker-Crane_1_Target_1_Position    | DInt     | 0         | False    | True                               | True                               | True                        | False        |             |           |  |
| Busy                                 | Bool     | false     | False    | True                               | True                               | True                        | False        |             |           |  |
| Ready                                | Bool     | false     | False    | True                               | True                               | True                        | False        |             |           |  |
| WarehouseFull                        | Bool     | false     | False    | True                               | True                               | True                        | False        |             |           |  |
| OccupiedPositions                    | DInt     | 0         | False    | True                               | True                               | True                        | False        |             |           |  |
| InOut                                |          |           |          |                                    |                                    |                             |              |             |           |  |
| ▼ Static                             |          |           |          |                                    |                                    |                             |              |             |           |  |
| State                                | Int      | 0         | False    | True                               | True                               | True                        | False        |             |           |  |
| NextPosition                         | DInt     | 0         | False    | True                               | True                               | True                        | False        |             |           |  |
| MovementStarted                      | Bool     | false     | False    | True                               | True                               | True                        | False        |             |           |  |
| ▼ TimerLiftUp                        | TON_TIME |           | False    | True                               | True                               | True                        | False        |             |           |  |
| PT                                   | Time     | T#0ms     | False    | True                               | True                               | True                        | False        |             |           |  |
| ET                                   | Time     | T#0ms     | False    | True                               | False                              | True                        | False        |             |           |  |
| IN                                   | Bool     | false     | False    | True                               | True                               | True                        | False        |             |           |  |
| Q                                    | Bool     | false     | False    | True                               | False                              | True                        | False        |             |           |  |
| ▼ TimerLiftDown                      | TON_TIME |           | False    | True                               | True                               | True                        | False        |             |           |  |
| PT                                   | Time     | T#0ms     | False    | True                               | True                               | True                        | False        |             |           |  |
| ET                                   | Time     | T#0ms     | False    | True                               | False                              | True                        | False        |             |           |  |
| IN                                   | Bool     | false     | False    | True                               | True                               | True                        | False        |             |           |  |
| Q                                    | Bool     | false     | False    | True                               | False                              | True                        | False        |             |           |  |
| ▼ TimerDeposit                       | TON_TIME |           | False    | True                               | True                               | True                        | True         |             |           |  |
| PT                                   | Time     | T#0ms     | False    | True                               | True                               | True                        | False        |             |           |  |
| ET                                   | Time     | T#0ms     | False    | True                               | False                              | True                        | False        |             |           |  |
| IN                                   | Bool     | false     | False    | True                               | True                               | True                        | False        |             |           |  |
| Q                                    | Bool     | false     | False    | True                               | False                              | True                        | False        |             |           |  |